

Transformation

Standard Transformation Matrix

EXAMPLE Suppose T is linear transformation from $\mathbb{R}^2$ into $\mathbb{R}^3$ such that

$$T(\mathbf{e}_1) = \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix} \quad \text{and} \quad T(\mathbf{e}_2) = \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix}$$

find a formula for the image of an arbitrary $\mathbf{x}$ in $\mathbb{R}^2$.

SOLUTION Write

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 \tag{1}$$

Standard Transformation Matrix

SOLUTION

(Continued...)

Since T is a *linear* transformation,

$$T(\mathbf{x}) = x_1 T(\mathbf{e}_1) + x_2 T(\mathbf{e}_2) \quad (2)$$

$$= x_1 \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix} = \begin{bmatrix} 5x_1 - 3x_2 \\ -7x_1 + 8x_2 \\ 2x_1 + 0 \end{bmatrix} \quad \blacksquare$$

The step from equation (1) to equation (2) explains why knowledge of $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$ is sufficient to determine $T(\mathbf{x})$ for any $\mathbf{x}$. Moreover, since (2) expresses $T(\mathbf{x})$ as a linear combination of vectors, we can put these vectors into the columns of a matrix A and write (2) as

$$T(\mathbf{x}) = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A\mathbf{x}$$

Standard Transformation Matrix

EXAMPLE Find the standard matrix A for the dilation transformation $T(\mathbf{x}) = 3\mathbf{x}$, for $\mathbf{x}$ in $\mathbb{R}^2$.

SOLUTION Write

$$T(\mathbf{e}_1) = 3\mathbf{e}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \quad \text{and} \quad T(\mathbf{e}_2) = 3\mathbf{e}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

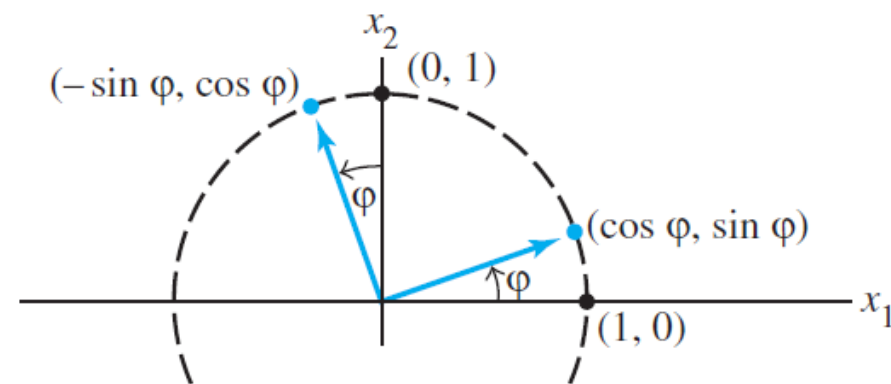
Standard Transformation Matrix

EXAMPLE 8 Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the transformation that rotates each point in $\mathbb{R}^2$ about the origin through an angle φ , with counterclockwise rotation for a positive angle. We could show geometrically that such a transformation is linear.

Find the standard matrix A of this transformation.

SOLUTION $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ rotates into $\begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ rotates into $\begin{bmatrix} -\sin \varphi \\ \cos \varphi \end{bmatrix}$.

$$A = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix}$$



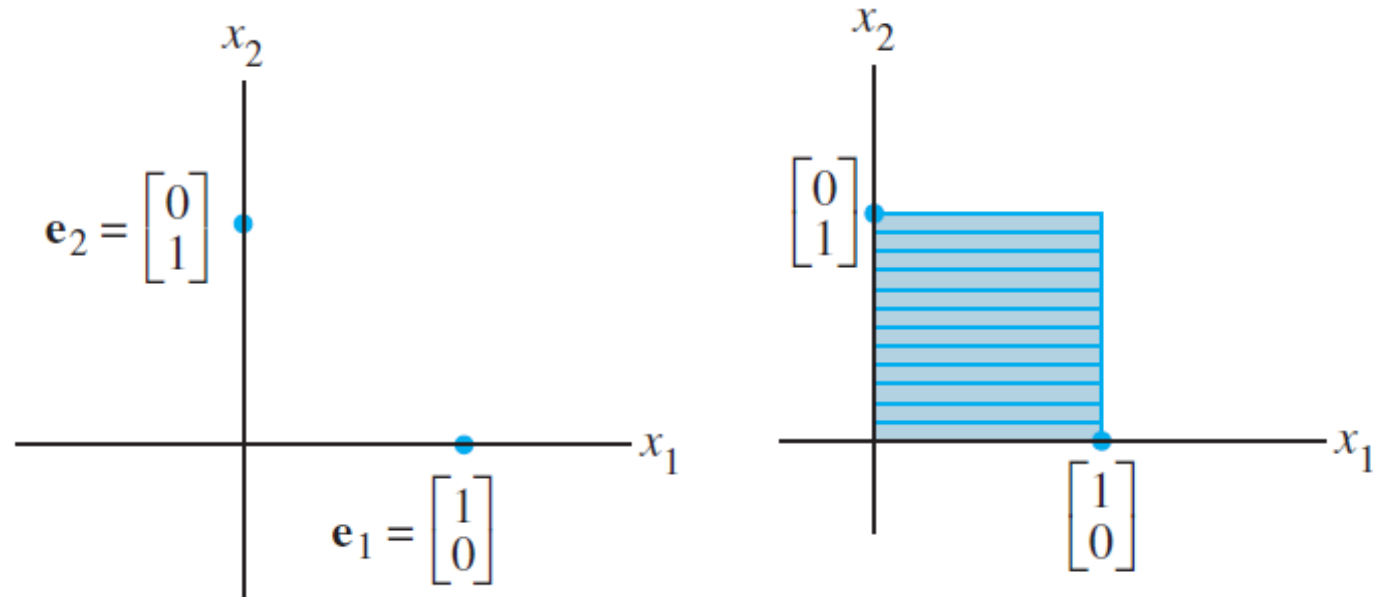
Geometric Linear Transformations of $\mathbb{R}^2$

Transformation of a Unit Square

- We can construct a unit square as shown in figure below, which is a 2×2 identity matrix I_2 .
- We can then observe what a transformation does to that unit square to get a better clarity.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$



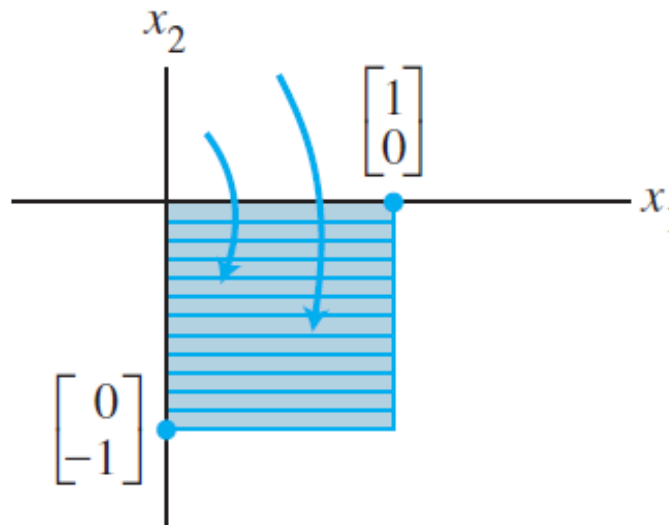
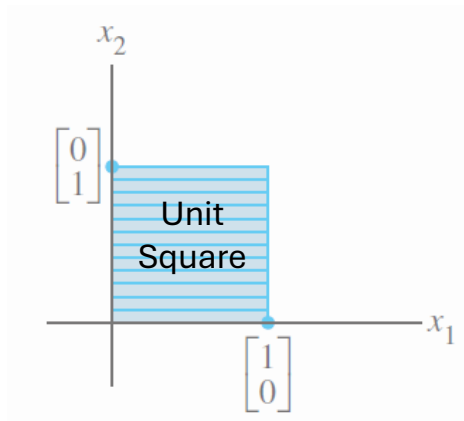
Geometric Linear Transformations of $\mathbb{R}^2$

Reflections (1/5)

Transformation	Image of the Unit Square	Standard Matrix
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Reflection through
the x_1 -axis

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



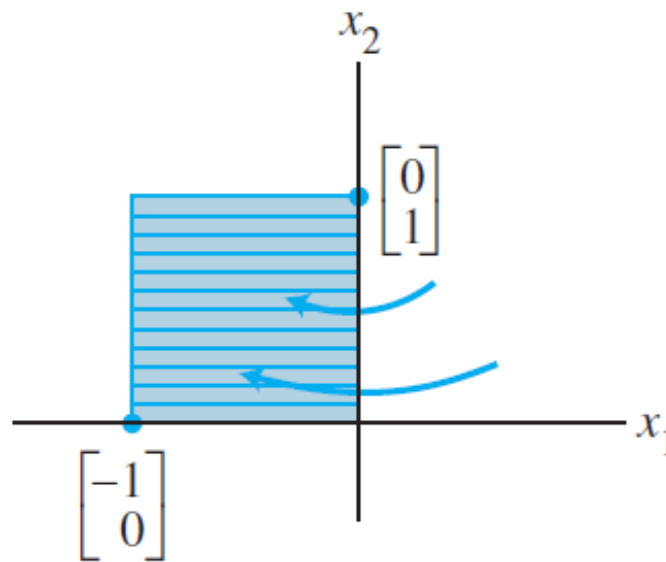
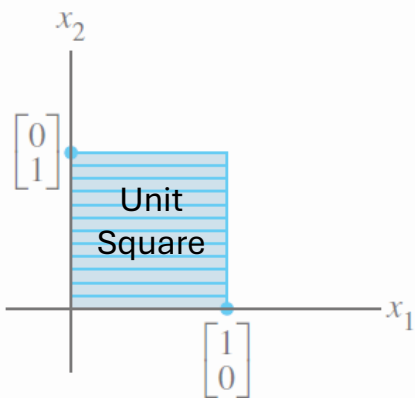
Geometric Linear Transformations of $\mathbb{R}^2$

Reflections (2/5)

Transformation	Image of the Unit Square	Standard Matrix
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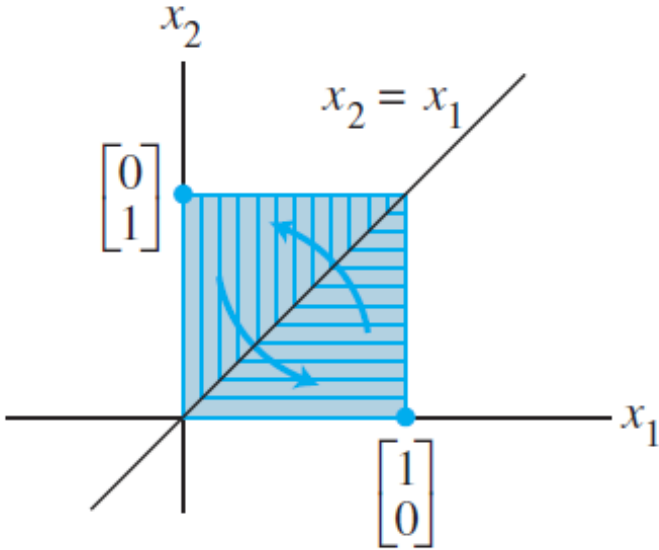
Reflection through
the x_2 -axis

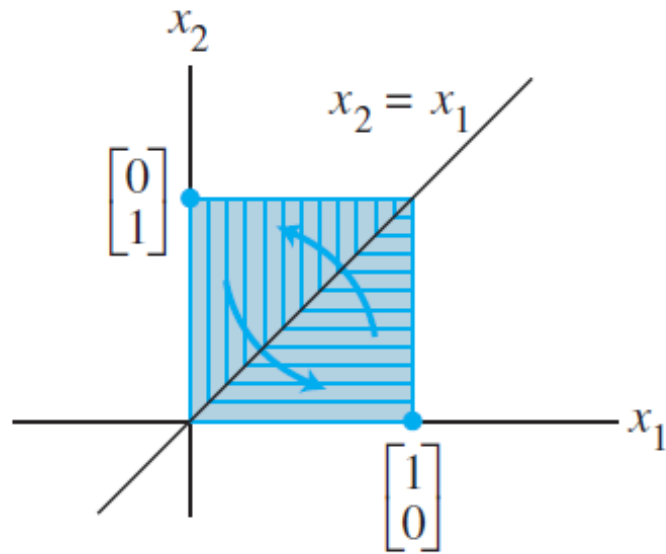
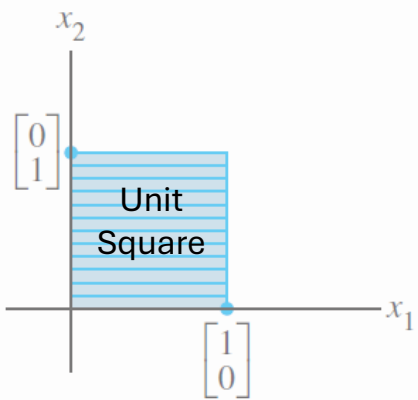
$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$



Geometric Linear Transformations of $\mathbb{R}^2$

Reflections (3/5)

Transformation	Image of the Unit Square	Standard Matrix
Reflection through the line $x_2 = x_1$		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$



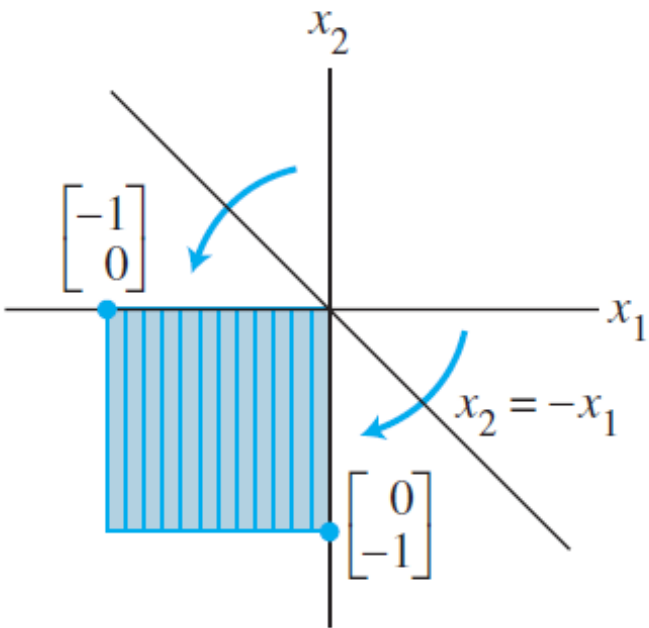
Geometric Linear Transformations of $\mathbb{R}^2$

Reflections (4/5)

Transformation	Image of the Unit Square	Standard Matrix
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Reflection through
the line $x_2 = -x_1$

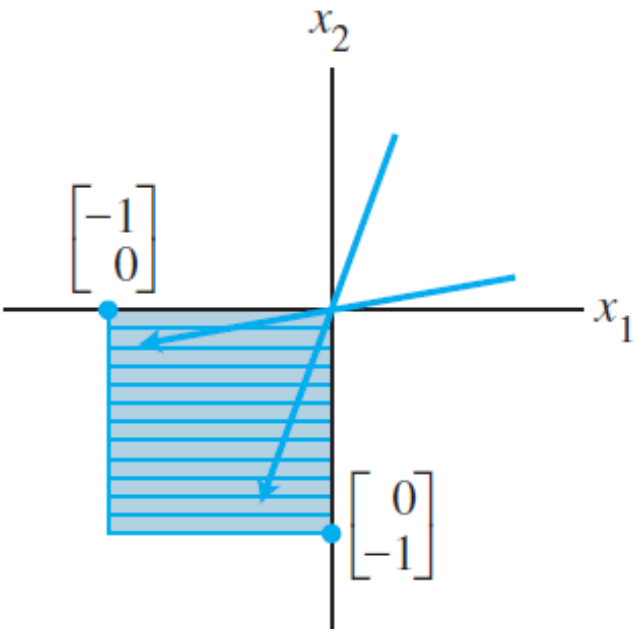
$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$



Geometric Linear Transformations of $\mathbb{R}^2$
Unit Square

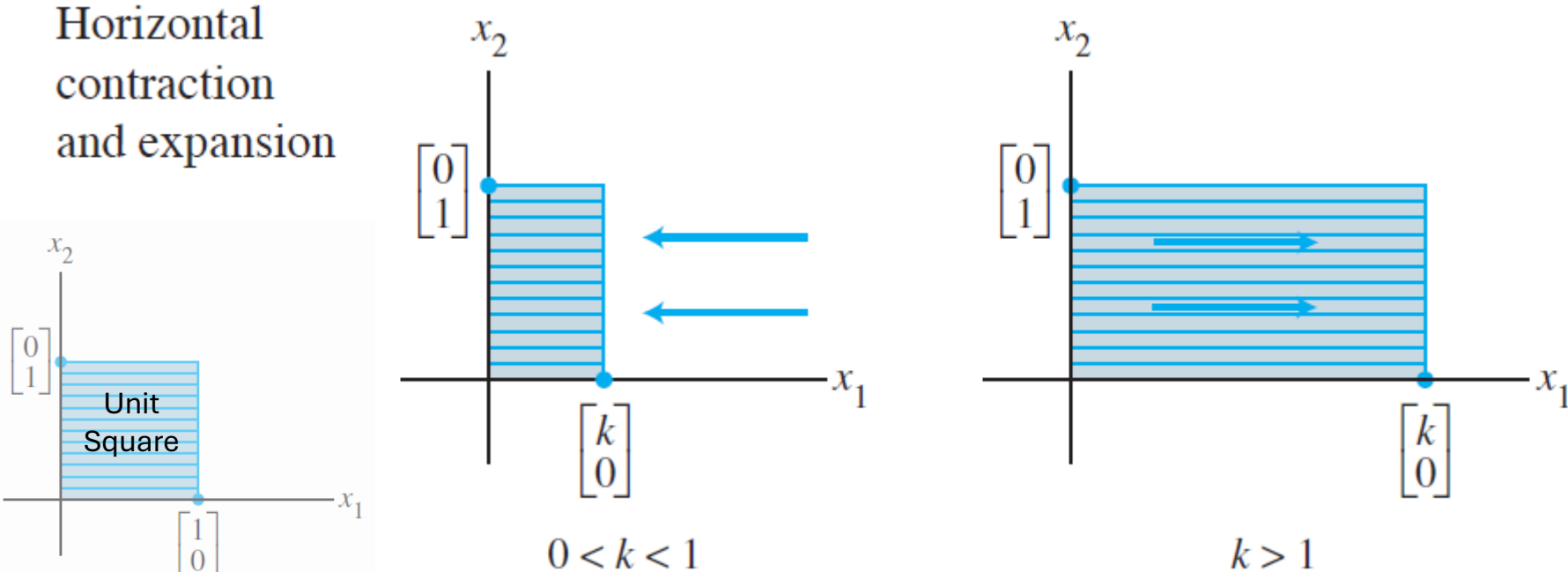
Geometric Linear Transformations of $\mathbb{R}^2$

Reflections (5/5)

Transformation	Image of the Unit Square	Standard Matrix
Reflection through the origin		$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

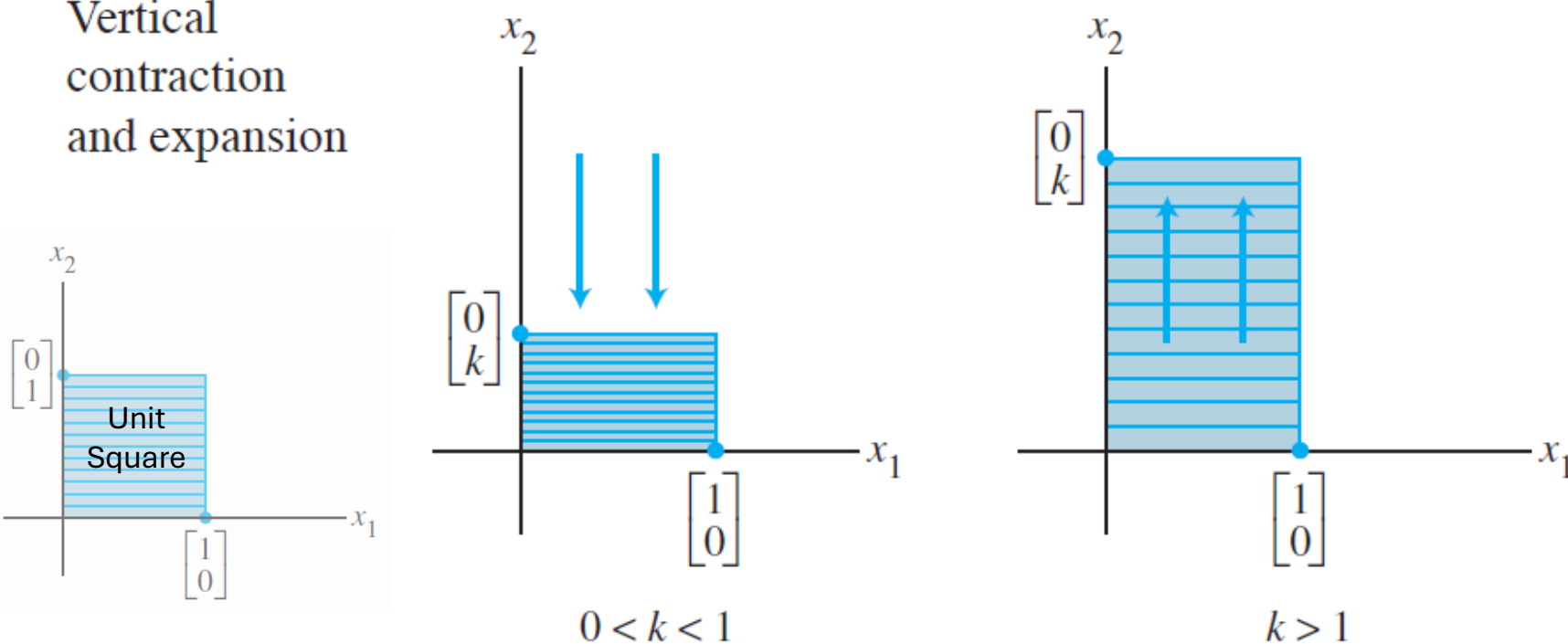
Geometric Linear Transformations of $\mathbb{R}^2$

Contractions and Expansions (1/2)

Transformation	Image of the Unit Square	Standard Matrix
Horizontal contraction and expansion	 Horizontal contraction and expansion Unit Square $0 < k < 1$ $k > 1$	$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$

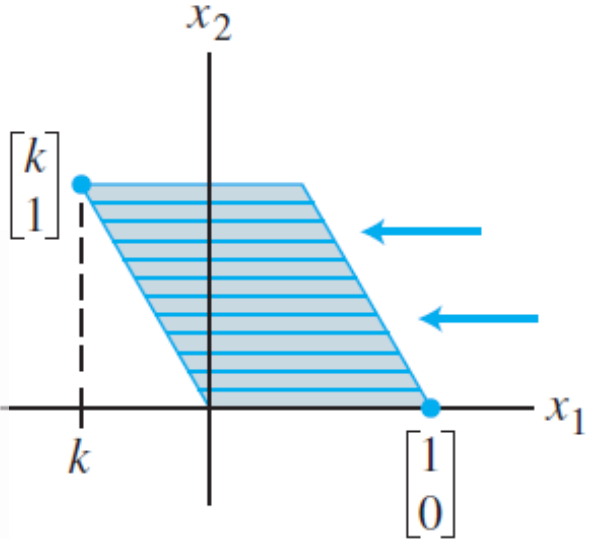
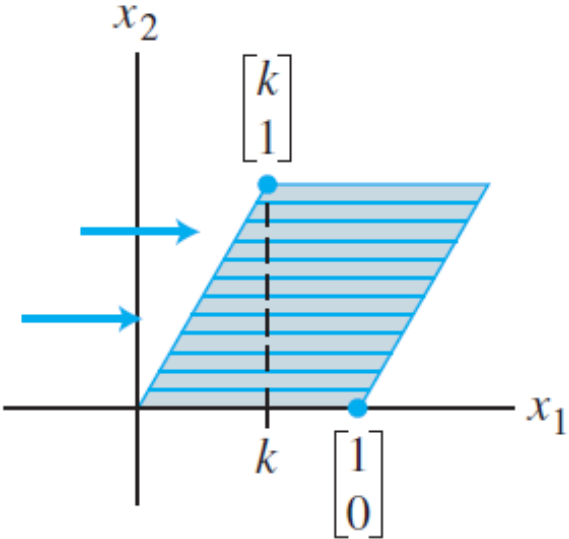
Geometric Linear Transformations of $\mathbb{R}^2$

Contractions and Expansions (2/2)

Transformation	Image of the Unit Square	Standard Matrix
Vertical contraction and expansion	 The diagram illustrates the vertical contraction and expansion of the unit square. On the left, the unit square is shown in the first quadrant of a coordinate system with axes x_1 and x_2. Its vertices are labeled as column vectors: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ at the top-left, $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ at the bottom-right, and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ at the top-right. The text "Unit Square" is inside. To the right, two cases are shown. The first case, labeled $0 < k < 1$, shows a contracted rectangle with height k. The top-left vertex is $\begin{bmatrix} 0 \\ k \end{bmatrix}$ and the bottom-right is $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Two blue arrows point downwards from the original top edge to the new top edge. The second case, labeled $k > 1$, shows an expanded rectangle with height k. The top-left vertex is $\begin{bmatrix} 0 \\ k \end{bmatrix}$ and the bottom-right is $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Two blue arrows point upwards from the original top edge to the new top edge.	$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$

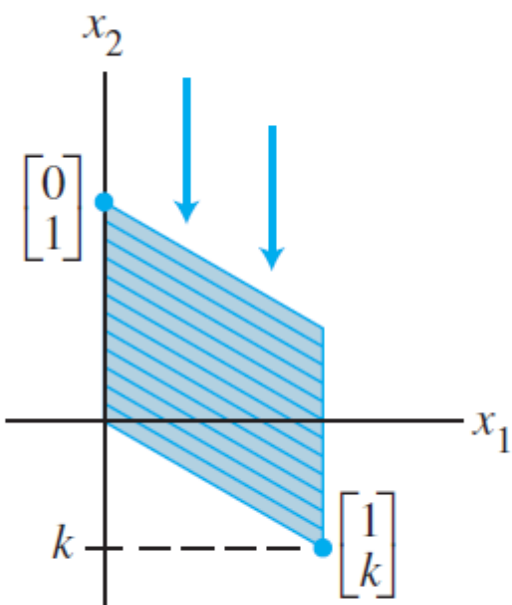
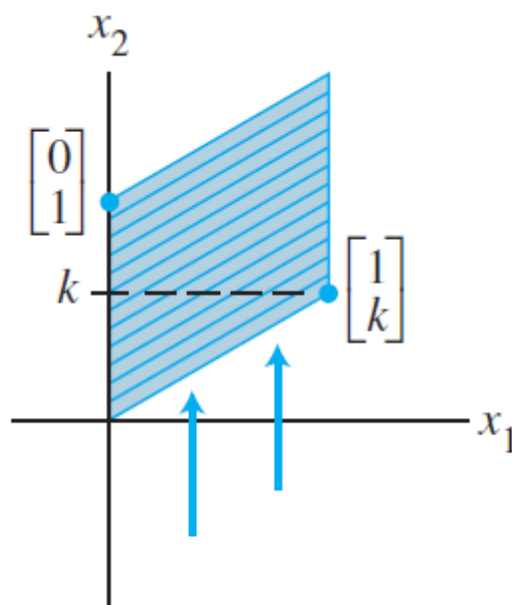
Geometric Linear Transformations of $\mathbb{R}^2$

Shears (1/2)

Transformation	Image of the Unit Square	Standard Matrix
Horizontal shear	 $k < 0$	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$
	 $k > 0$	

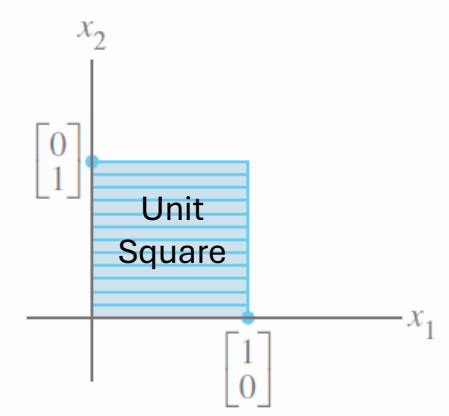
Geometric Linear Transformations of $\mathbb{R}^2$

Shears (2/2)

Transformation	Image of the Unit Square	Standard Matrix
Vertical shear	  $k < 0$ $k > 0$	$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$

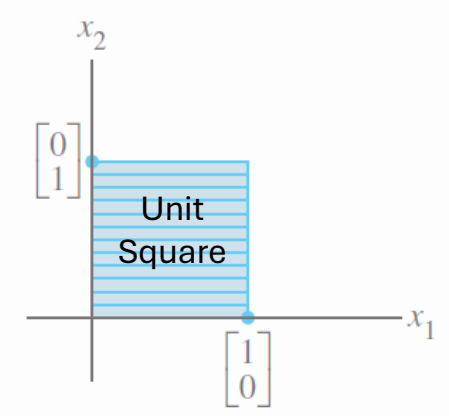
Geometric Linear Transformations of $\mathbb{R}^2$

Projections (1/2)

Transformation	Image of the Unit Square	Standard Matrix
Projection onto the x_1 -axis		$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Geometric Linear Transformations of $\mathbb{R}^2$

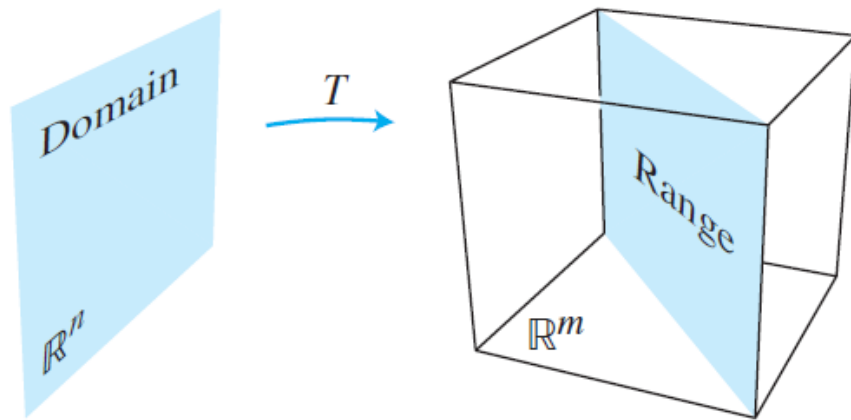
Projections (2/2)

Transformation	Image of the Unit Square	Standard Matrix
Projection onto the x_2 -axis		$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

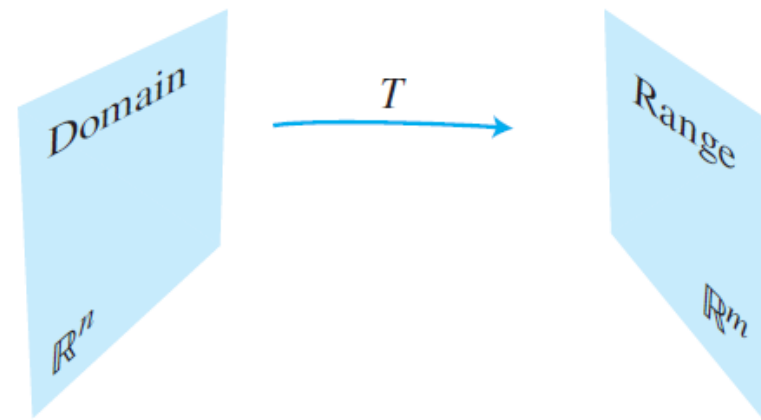
One-to-one Linear Transformations

Does T map $\mathbb{R}^n$ onto $\mathbb{R}^m$? (Does a solution exist?)

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **onto** $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at least one* $\mathbf{x}$ in $\mathbb{R}^n$.



T is not onto $\mathbb{R}^m$



T is onto $\mathbb{R}^m$

One to One

Here are some equivalent ways of saying that T is one-to-one:

- For every vector b in $\mathbf{R}^m$, the equation $T(x) = b$ has *zero or one* solution x in $\mathbf{R}^n$.
- Different inputs of T have different outputs.
- If $T(u) = T(v)$ then $u = v$.

Not One to One

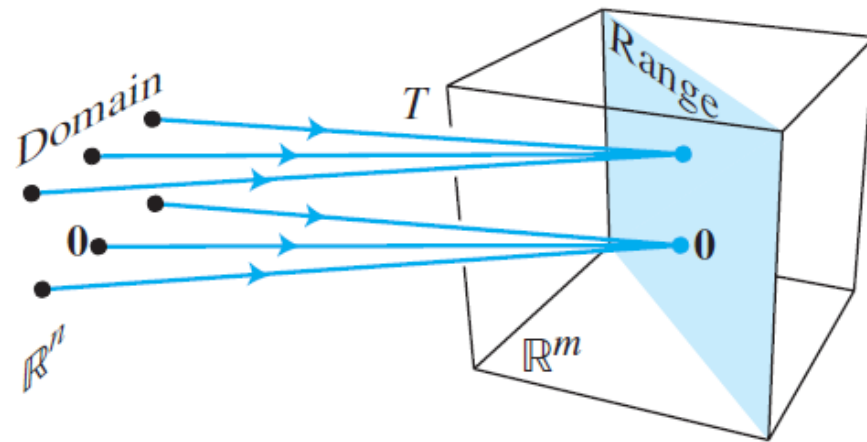
Here are some equivalent ways of saying that T is *not* one-to-one:

- There exists some vector b in $\mathbf{R}^m$ such that the equation $T(x) = b$ has *more than one* solution x in $\mathbf{R}^n$.
- There are two different inputs of T with the same output.
- There exist vectors u, v such that $u \neq v$ but $T(u) = T(v)$.

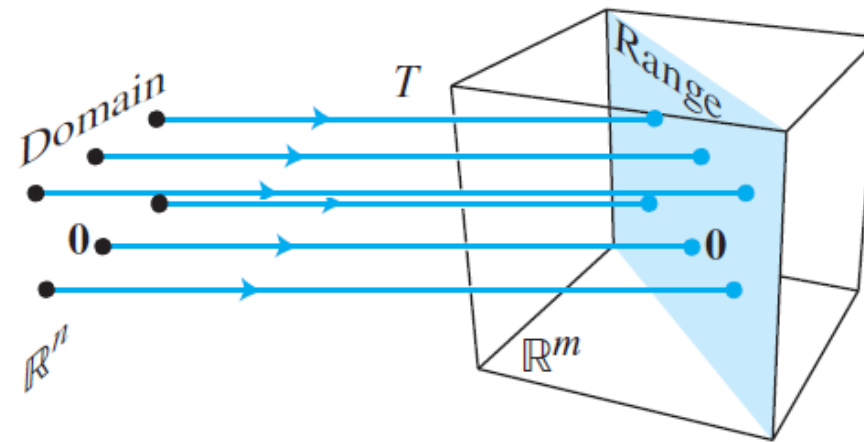
One-to-one Linear Transformations

One-to-one mapping (Is it a unique solution?)

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **one-to-one** if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at most one* $\mathbf{x}$ in $\mathbb{R}^n$.



T is not one-to-one



T is one-to-one

Is every $\mathbf{b}$ the image of at most one vector?

One-to-one Linear Transformations

Theorem

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation and let A be the standard matrix for T . Then:

- a. T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if the columns of A span $\mathbb{R}^m$;
- b. T is one-to-one if and only if the columns of A are linearly independent.

A one-to-one linear transformation (also called an injective transformation) in linear algebra is a transformation where each vector in the domain is mapped to a unique vector in the codomain. In other words, no two distinct vectors in the domain map to the same vector in the codomain.

To be **one-to-one**, a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ must satisfy:

$$T(\mathbf{u}) = T(\mathbf{v}) \implies \mathbf{u} = \mathbf{v}$$

This is equivalent to saying that the **kernel** of the transformation contains only the zero vector, i.e.,

$$T(\mathbf{v}) = \mathbf{0} \implies \mathbf{v} = \mathbf{0}.$$

Example

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$$

The transformation T takes a vector $\mathbf{v} \in \mathbb{R}^2$ and maps it to $A\mathbf{v}$.

Step 1: Verify if T is One-to-One

To determine if T is one-to-one, we need to check whether the matrix A is invertible. If A is invertible, then T is one-to-one.

The invertibility of a matrix can be checked by calculating its **determinant**. If the determinant is non-zero, the matrix is invertible, and the transformation is one-to-one.

The determinant of A is:

$$\det(A) = (2)(3) - (1)(1) = 6 - 1 = 5$$

Since $\det(A) = 5 \neq 0$, the matrix A is invertible, and therefore the transformation T is one-to-one.